# Part 1:
# Development of a short course:
# Introduction to Statistical inference and estimation using R

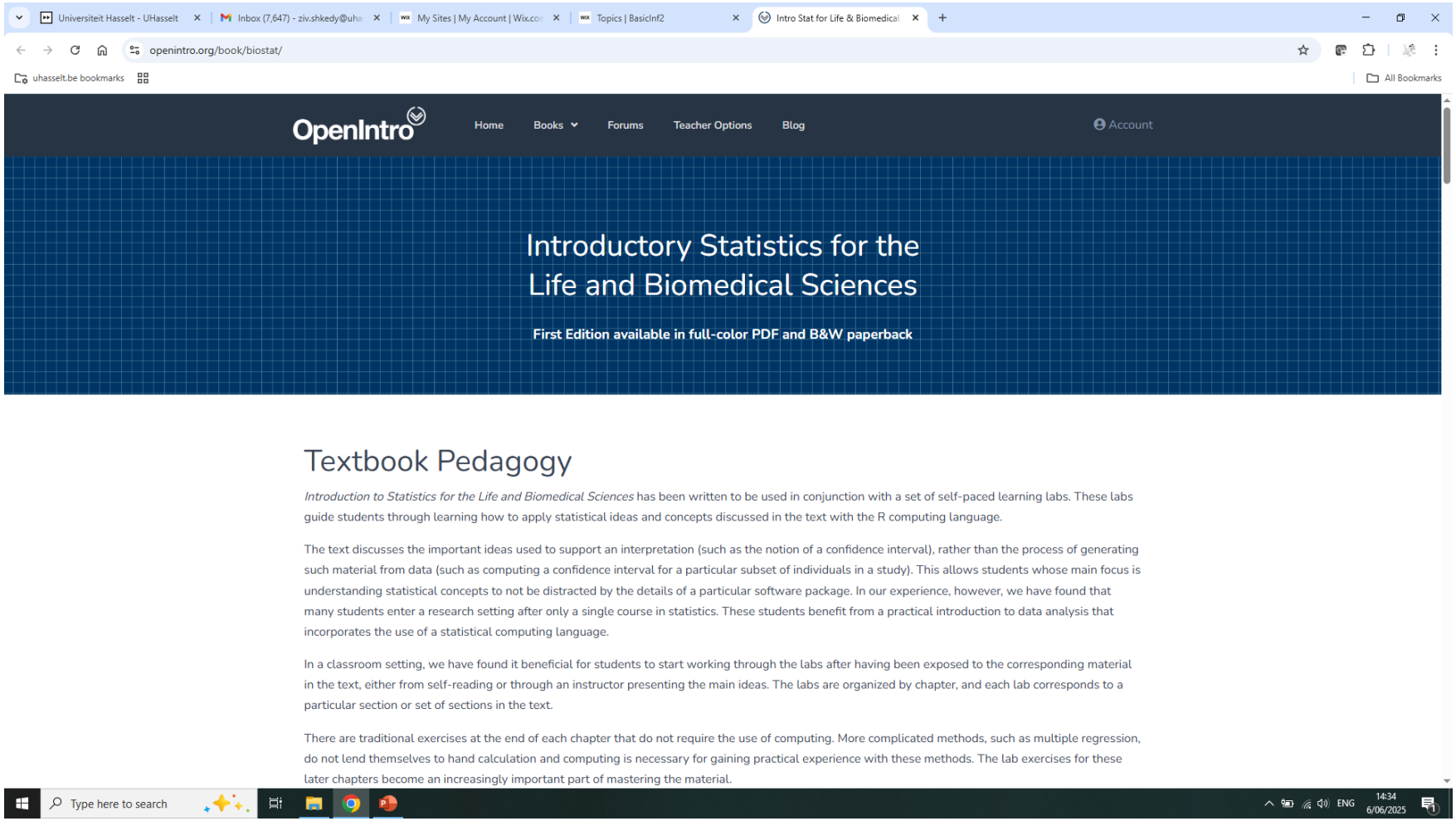
# Course materials

- The following materials are available for the course:
  - Book (in PDF).
  - Slides (PDF/PP).
  - R program for all the examples presented in the slides.
  - Online book.
  - Course website.

# Topics that we cover today

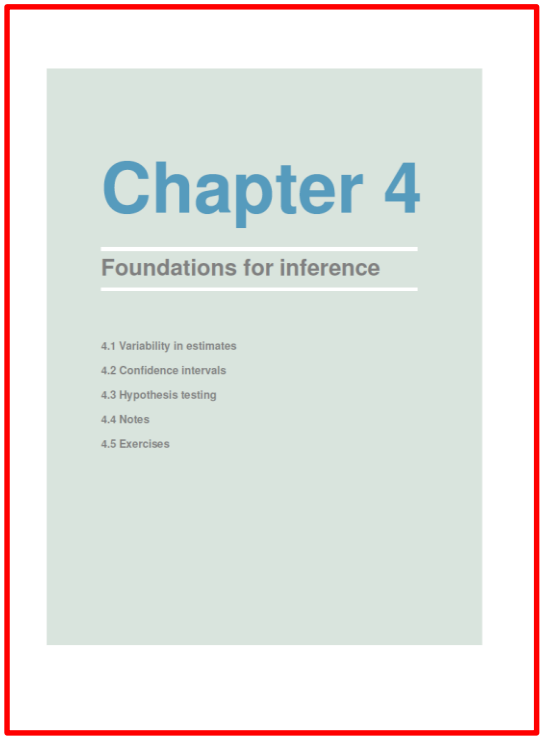
- Starting point: online book (in PDF format).
- The short course covers one chapter from the book.
- Main focus today:
  - How to create content based on the book chapter.

# An online (free) book




**https://www.openintro.org/book/biostat/**

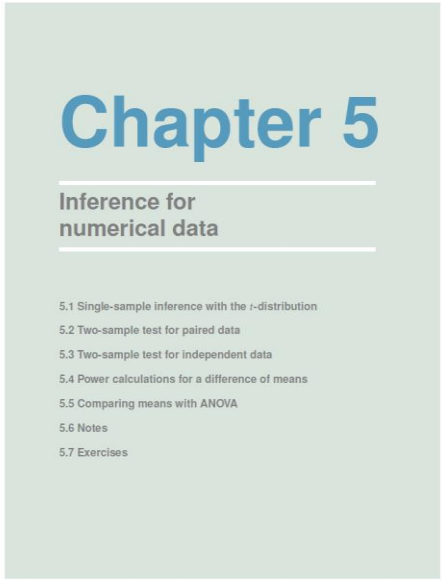
# Three chapters are related to inference



## Chapter 4

### Foundations for inference

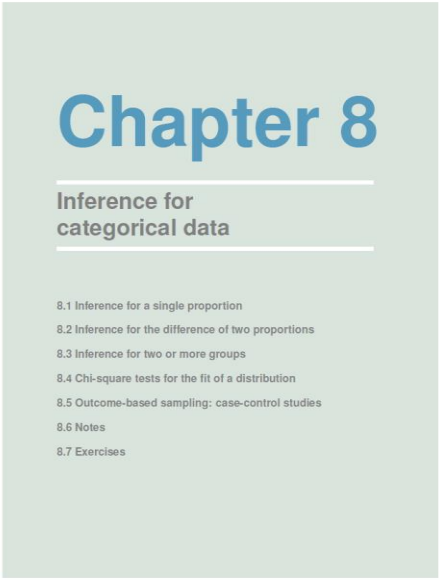

4.1 Variability in estimates
4.2 Confidence intervals
4.3 Hypothesis testing
4.4 Notes
4.5 Exercises

## Chapter 5

### Inference for numerical data


5.1 Single-sample inference with the $t$-distribution
5.2 Two-sample test for paired data
5.3 Two-sample test for independent data
5.4 Power calculations for a difference of means
5.5 Comparing means with ANOVA
5.6 Notes
5.7 Exercises

## Chapter 8

### Inference for categorical data


8.1 Inference for a single proportion
8.2 Inference for the difference of two proportions
8.3 Inference for two or more groups
8.4 Chi-square tests for the fit of a distribution
8.5 Outcome-based sampling: case-control studies
8.6 Notes
8.7 Exercises


- The short course that we develop is focused on Chapter 4.

# Course level

- Book chapter (PDF): “Light theory” + examples.
- Online book (HTML): the book chapter + extra examples in R.

# For whom ?

- (Bio)Stat + data science: under graduate students and master students (for the practical examples in R).

- Non statisticians: all levels.

# Short description of the course

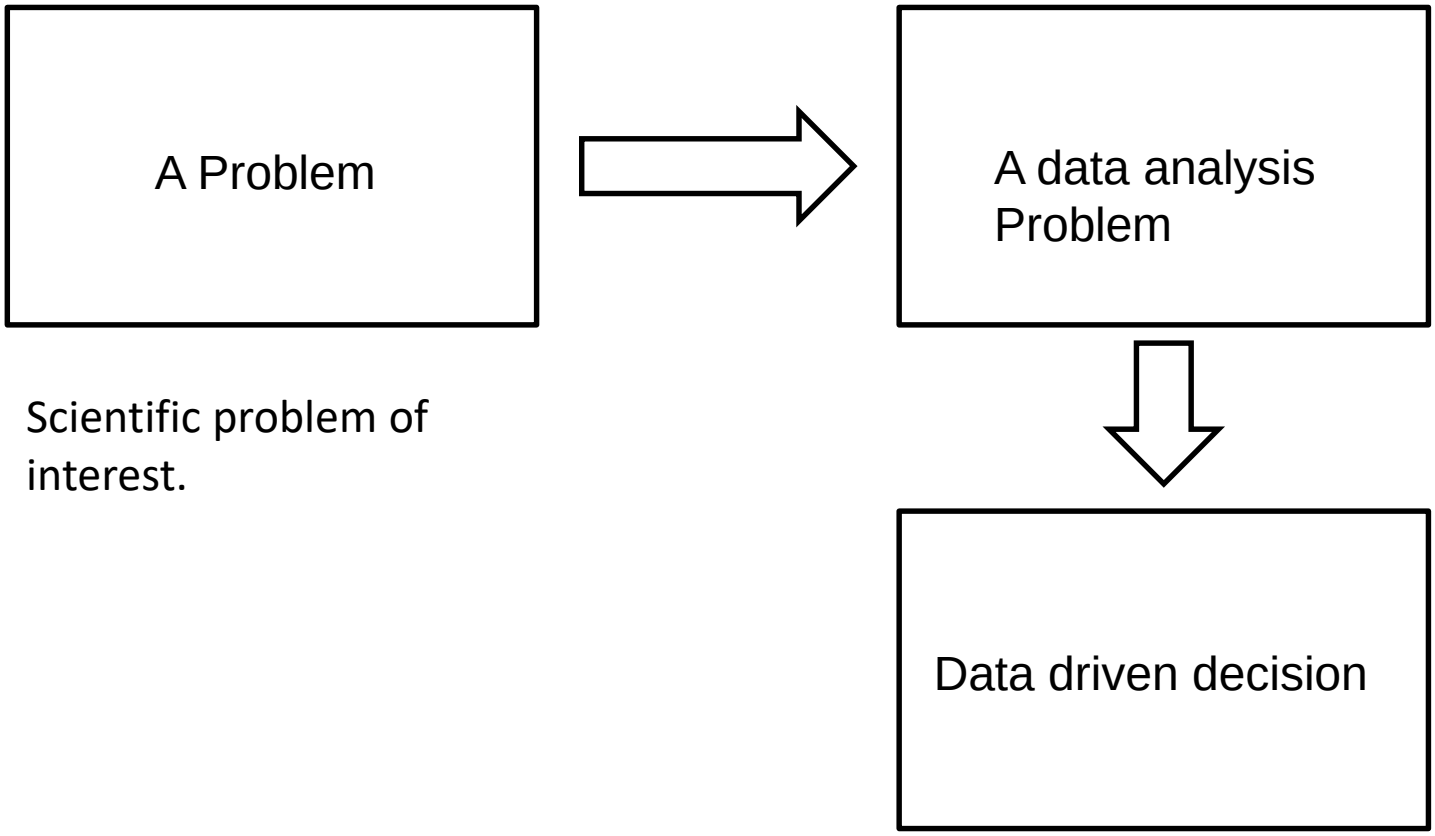
# Course structure

- Introduction: steps in data analysis
- Analysis of numerical data: point estimators and interval estimates
- Visualization tool
- Interval estimates.
- Inference for numerical data for one population
- Inference for numerical data for one/two population(s)

# Steps in data analysis

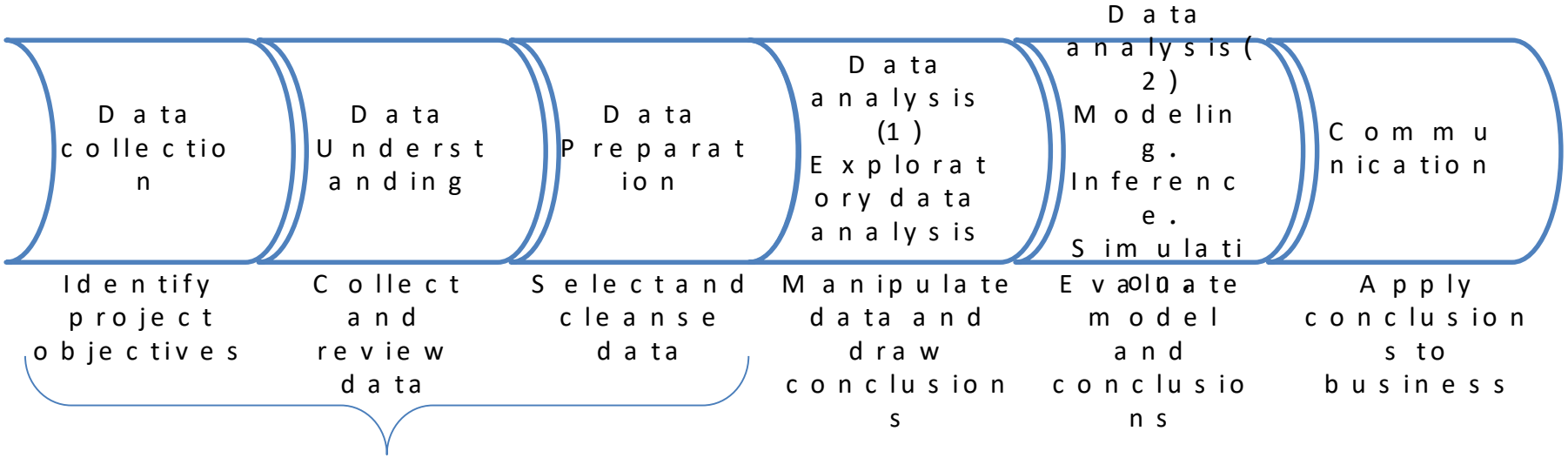
# Steps in data analysis

- Data analysis approach in the course:



# Steps in data analysis

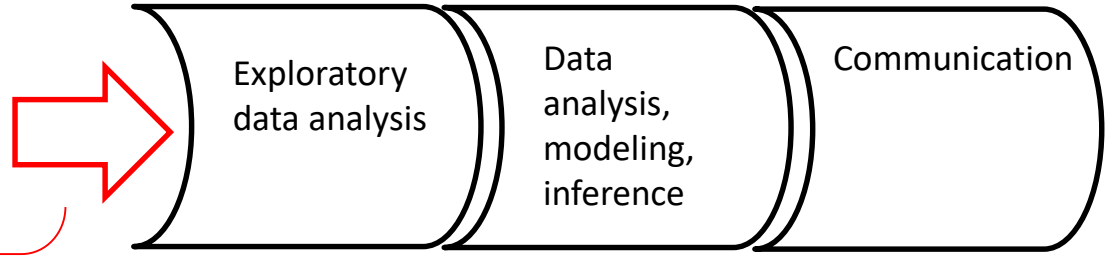
- Steps related to data analysis:




Not a part of our course

# Steps in data analysis

Estimating the mean BMI in the population.




Scientific problem of interest: how to estimate the mean BMI?

Methodology: inference for one sample:

- Point estimates.
- Interval estimates.
- Hypothesis testing.

A report.

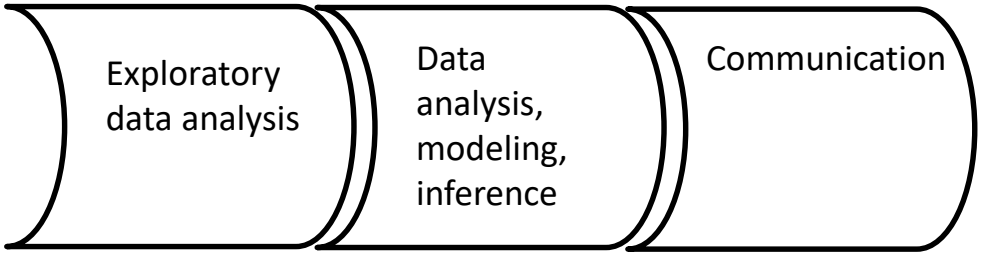
The solution

# Steps in data analysis

Estimating the mean BMI in the population.

Scientific problem of interest: how to model the association ?

Methodology: inference for one sample.




Histogram, Boxplot…

Confidence intervals and test of hypothesis.

A report.

The solution

# Steps in data analysis

Estimating the mean BMI in the population.

Scientific problem of interest: how to estimate the mean BMI?

Methodology: inference for one sample.

We "translate" the methodology to software usage

Exploratory data analysis

Data analysis, modeling, inference

Communication

Histogram, Boxplot…

Confidence intervals and test of hypothesis.

A report.

ggplot2()

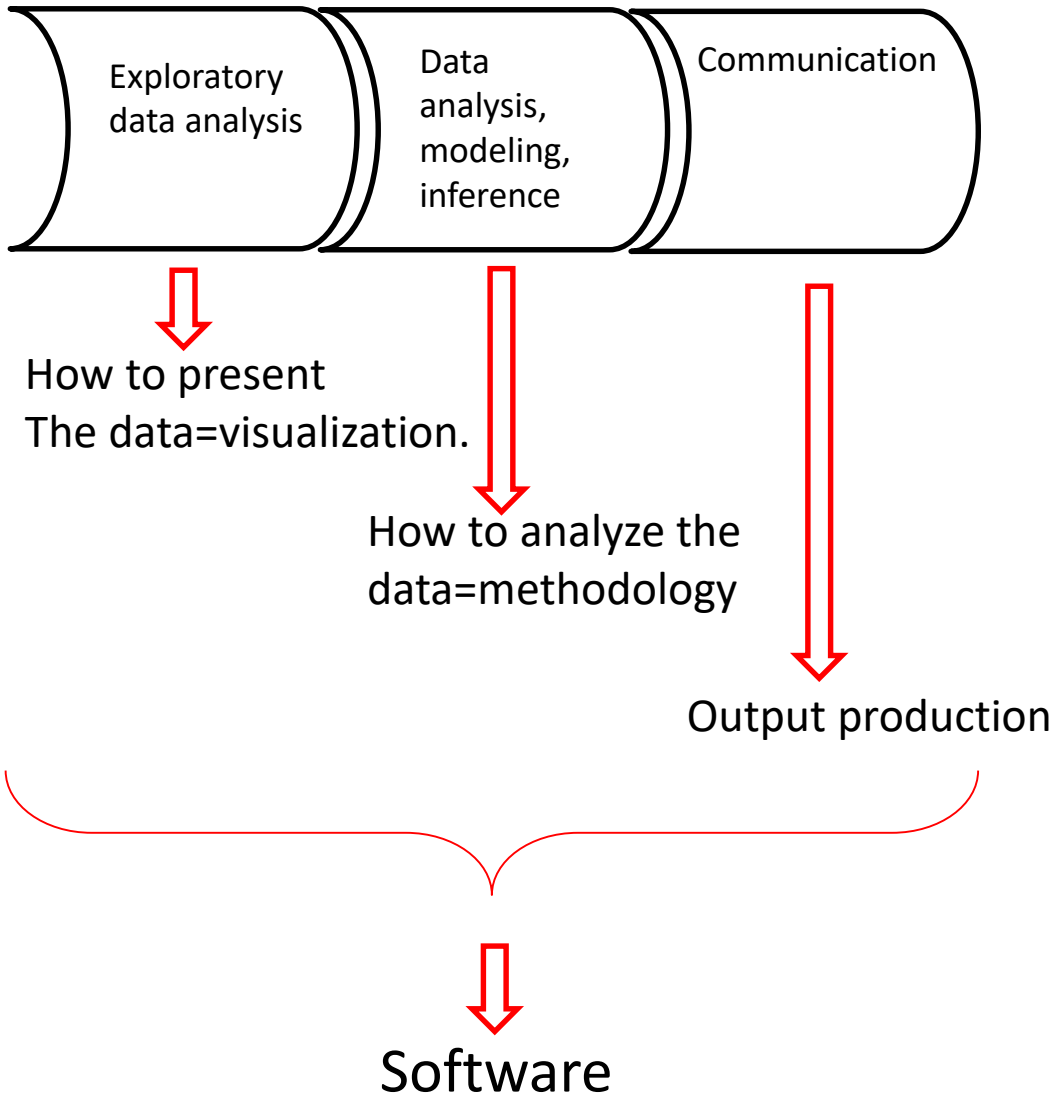
z.test ()

R markdown to produce a HTML file.

**We develop software to produce the solution and to communicate the solution**

# How was the course developed ?

# Steps in data analysis



# Software

# Software

- All examples: illustrated in R.
- You need to install:
  - R.
  - R-Studio.
- R packages for the course:
  - R markdown.
  - z.test().
  - ggplot2.
  - …